

HOME ASSIGNMENT

Classification Algorithm

(a) age < 30 & gov less & urban

Age - 23.0 yrs

Govt. satisfaction - 0.400 /10

Urban voter - Yes

Output in model:

Party A:

$$0.000 \div 7100 = 0.0\%$$

Party T:

$$7095 \div 7100 = 99.9\%$$

Party C:

$$3.32 \div 7100 = 0.0\%$$

Undecided:

$$1.37 \div 7100 = 0.0\%$$

$$0.000 + 7095 + 3.32 + 1.37 = 7099.69 \approx 7100$$

My Calculation:

Party A:

$$0.000015 \div 7131.71 \times 100 \approx 0.0000002\%$$

Party T:

$$7127 \div 7131.71 \times 100 \approx 99.93\%$$

Party C:

$$3.33 \div 7131.71 \times 100 \approx 0.047\%$$

Undecided:

$$1.38 \div 7131.71 \times 100 \approx 0.019\%$$

$$0.000015 + 7127 + 3.33 + 1.38 \approx 7131.71$$

Party T is correct. The final prediction Party T and its approximately 99.9% share are correct.

Step 2b — each class's share = its own e^{score} ÷ that same total (7100)

Party A $0.000 \div 7100 = 0.0\%$

Party T $7095 \div 7100 = 99.9\%$

Party C $3.32 \div 7100 = 0.0\%$

Undecided $1.37 \div 7100 = 0.0\%$

(b) age > 30 & gov less & non urban

Age - 38.0 yrs

Govt. satisfaction - 0.400 /10

Urban voter - No

Output in model:

Step 2b — each class's share = its own e^{score} ÷ that same total (650)

Party A $0.000 \div 650 = 0.0\%$

Party T $558 \div 650 = 85.9\%$

Party C $89.8 \div 650 = 13.8\%$

Undecided $2.22 \div 650 = 0.3\%$

My Calculation:

Party A:

$$e^{-9.267} \approx 0.000094$$

$$\div \approx 647.7 \times 100 \approx 0.000015\%$$

Party T:

$$e^{6.322} \approx 555.5$$

$$\div \approx 647.7 \times 100 \approx 85.76\%$$

Party C:

$$e^{4.502} \approx 90.2$$

$$\div \approx 647.7 \times 100 \approx 13.93\%$$

Undecided:

$$e^{0.798} \approx 2.22$$

$$\div \approx 647.7 \times 100 \approx 0.34\%$$

$$0.000094 + 555.5 + 90.2 + 2.22 \approx 647.92$$

Party T is correct. The final prediction Party T and its approximately 85.8% share are correct.

Linear Algorithm

(a)QA Salary

Experience < 9 & skill rating more

Experience - 5.50 yrs

Skill rating - 7.00 /10

Output in Model:

Base salary — the starting value, before any feature counts

3.10 LPA

Experience

0.966 LPA per year $\times$ 5.50 yrs

= +5.32 LPA

Skill rating

1.34 LPA per point $\times$ 7.00 /10

= +9.39 LPA

TOTAL 17.8 LPA

My Calculation:

Skill rating:

$$1.34 \times 7.00 = 9.38 \text{ LPA, not } 9.39 \text{ LPA}$$

$$3.10 + 5.313 + 9.38 = 17.793 \text{ LPA} \approx 17.79 \text{ LPA} \approx 17.8 \text{ LPA}$$

The final total of 17.8 LPA is correct after rounding, but the displayed Skill rating contribution should be 9.38 LPA, not 9.39 LPA.

Experience > 9 & skill rating less

Experience - 13.0 yrs

Skill rating - 3.00 /10

Output in Model:

Base salary — the starting value, before any feature counts

3.10 LPA

Experience

0.966 LPA per year $\times$ 13.0 yrs

= +12.6 LPA

Skill rating

1.34 LPA per point $\times$ 3.00 /10

= +4.02 LPA

TOTAL 19.7 LPA

My Calculation:

Same, hence no error.

(b)Food Delivery Time

Distance < 5 & traffic more

Distance - 4.00 km

Traffic - 6.00 /10

Output in Model:

Base time — the starting value, before any feature counts

7.56 min

Distance

$$2.40 \text{ min per km} \times 4.00 \text{ km}$$

$$= +9.61 \text{ min}$$

Traffic

$$3.16 \text{ min per point} \times 6.00 / 10$$

$$= +18.9 \text{ min}$$

TOTAL 36.1 min

My Calculation:

Distance:

$$2.40 \times 4.00 = 9.60 \text{ min, not 9.61}$$

$$7.56 + 9.60 + 18.96 = 36.12 \text{ min} \approx 36.1 \text{ min}$$

Hence final total of 36.1 min is still correct after rounding.

Distance > 5 & traffic less

Distance - 22.0 km

Traffic - 2.60 /10

Output in Model:

Base time — the starting value, before any feature counts

7.56 min

Distance

$$2.40 \text{ min per km} \times 22.0 \text{ km}$$

$$= +52.8 \text{ min}$$

Traffic

$$3.16 \text{ min per point} \times 2.60 / 10$$

$$= +8.21 \text{ min}$$

TOTAL 68.6 min

My Calculation:

Distance:

$2.40 \times 4.00 = 9.60$ min, not 9.61

$7.56 + 9.60 + 18.96 = 36.12$ min ≈ 36.1 min

Final total of 36.1 min is still correct after rounding.
